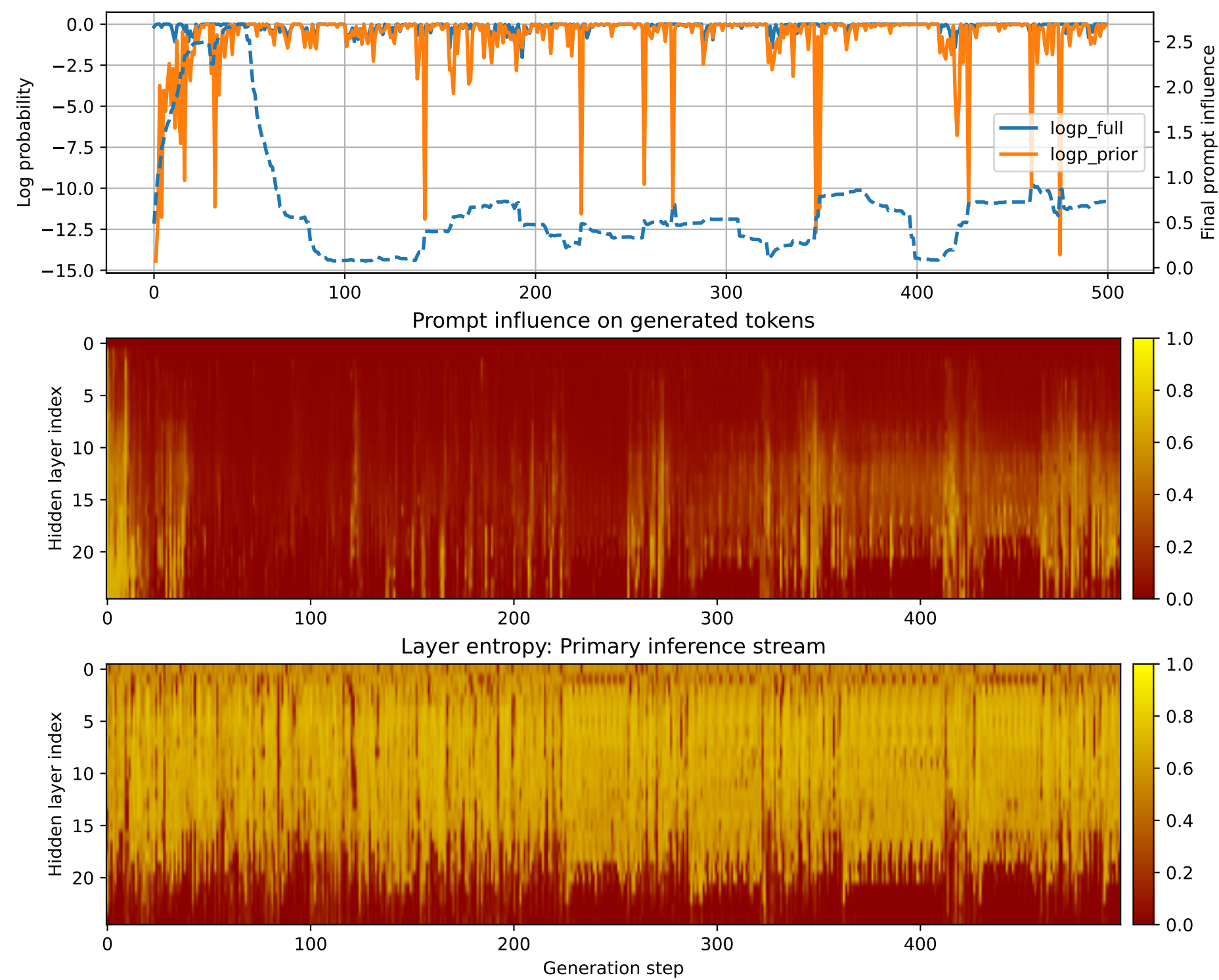


Prompt 1



PROMPT:
Task ID: PYC094 Return ONLY valid Python code. Do not use
Markdown fences. Do not include prose,
explanations, examples, print statements, or tests. Your code
must contain this exact check-token line: #
CHECK_TOKEN: PYC094 Your code must contain this required symbol:
def detect_anomalies_zscore Implement this
task: write function detect_anomalies_zscore(xs, threshold=3.0)..
Start your answer with the check-token
line, then provide the Python implementation.

MODEL RESPONSE:
CHECK_TOKEN: PYC094 # Function to detect anomalies in z-score
data using the z-score method def
detect_anomalies_zscore(xs, threshold=3.0): """ Detect
anomalies in z-score data using the z-score
method. Args: xs (list): A list of z-score values
for a dataset. threshold (float): The
threshold value for identifying anomalies. Defaults to 3.0.
Returns: list: A list of anomaly
scores, where each score indicates the z-score of a data point
above or below the threshold. """ #
Calculate z-scores for each data point z_scores = [(x -
mean(xs)) / std(xs) for x in xs] # Define a
function to determine if a data point is an anomaly def
is_anomaly(x): return x > threshold #
Apply the z-score threshold to each z-score and determine if it
is an anomaly anomalies = [x for x in
z_scores if is_anomaly(x)] return anomalies # Example usage
xs = [1, 2, 3, 4, 5, 6, 7, 8, 9, 10]
threshold = 3.0 anomalies = detect_anomalies_zscore(xs,
threshold) print(anomalies) # Output: [4.0, 5.0, 6.0,
7.0, 8.0, 9.0, 10.0] # Example usage with a different threshold
value threshold = 2.0 anomalies =
detect_anomalies_zscore(xs, threshold) print(anomalies) #
Output: [1.0, 2.0, 3.0, 4.0, 5.0, 6.0, 7.0, 8.0,
9.0, 10.0] # Example usage with a list of non-anomalous data
points xs = [1, 2, 3, 4, 5, 6, 7, 8, 9, 10]
threshold = 2.0 anomalies = detect_anomalies_zscore(xs,
threshold) print(anomalies) # Output: [1.0, 2.0,